

How do humans learn?

Sociocultural & Socio-political Foundations

SYS 478. Fall 2026. Prof. Sarah Van Wart



Today's Focus: "Context"

Learning does not happen in isolation. Context matters. "Context can refer to":

1. **Cultural contexts** – language, tools, relationships, communities, and shared practices.
2. **Socio-political contexts** – power, access, institutions, inequality, whose knowledge counts

Today's question: What does a social and political lens reveal that biology and psychology miss?

What is “Common Sense” / Tacit Knowledge?

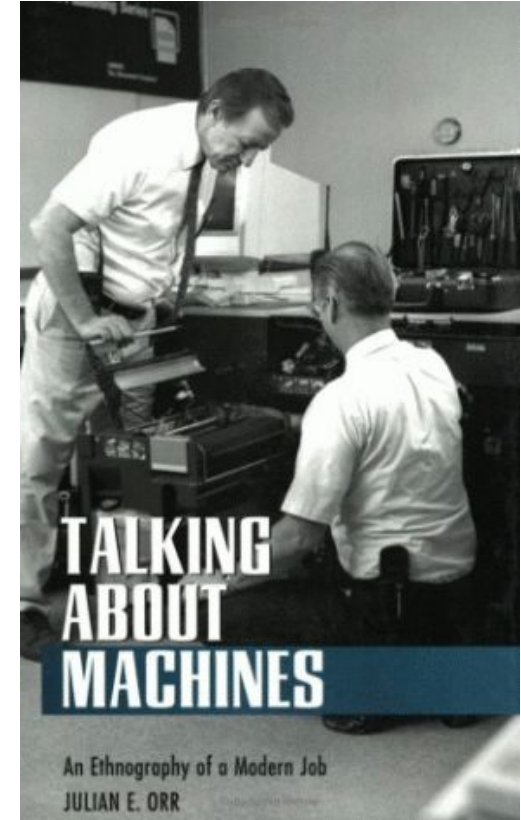
The background knowledge we use automatically without usually having to explain it.

- Some knowledge is hard to write down explicitly.
- People often learn by watching, practicing, and participating.
- Much of what counts as competence is informal.
- This is one reason instructions alone are often not enough.

Examples of Tacit Knowledge

Examples:

- Julian Orr and the Xerox Machine Instructions
- **“Work to rule” strikes:** A strike where workers only do what is explicitly stated in their job description, performing no additional tasks beyond the minimum required, effectively slowing down operations to pressure the employer into concessions.



Discuss: Learning “Common Sense”

- What is a practice or social cue that you had to learn without being explicitly taught?
- Could you have learned it via explicit instructions that you read or that were told to you?
- Think of a time you learned a social norm by breaking it. What happened?

Sociocultural Theory

- Learning is shaped by culture and history.
- What people learn depends in part on what their communities value.
- How people learn depends on available tools, norms, and relationships.
- Knowledge is often co-constructed rather than simply absorbed.

Lev Vygotsky: Influential Sociocultural Theorist

- Known for emphasizing the importance of the social and cultural dimensions of learning (tools, symbols, other people).
- **Zone of Proximal Development (ZPD)** – children learn best when they work on tasks slightly beyond their current abilities with guidance from more knowledgeable others.
- Developed many of his theories working with people with disabilities.



Example: Learning to Tie One's Shoes

A child first learns to tie their shoes with an adult saying, “cross the laces, make a loop, wrap it around...” Later, the child can guide themselves through the steps without help.

**First, the guidance is social. Then it becomes
part of the learner's own thinking.**

Luis Moll: Funds of Knowledge

- Households and communities contain valuable knowledge, skills, and practices the students have learned and bring with them into the classroom.
- Good teaching builds on what learners already know from family, work, language, and community life.
- The problem is not that students lack knowledge. The problem is often that schools fail to recognize the knowledge they already have.



Kris Gutiérrez: Re-Mediation & Tools

- People are ***made smart*** through the tools and resources they have access to in their environment.
- Being smart is relative and situated.

The question is not “How can we fix” Judith?” The question is: How do we reconfigure the learning environment – with resources, tools, and other forms of assistance – so that Judith can be the smart person that she already is?”



Kris Gutiérrez, UC Berkeley

Sociocultural Theory > Apprenticeship

- Definition: A model of learning involving guided participation in a community of practice (Jean Lave).
- Key Aspects:
 - Learning through observation and hands-on experience.
 - Mentorship and expert guidance.
- Examples: [apprenticeships](#) and internships.

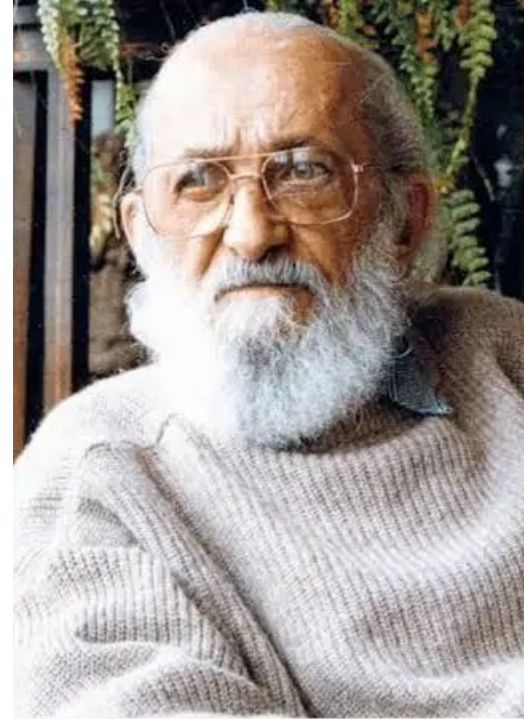


Socio-Political Considerations

(power dynamics)

Paulo Freire: Socio-Political Factors

- Brazilian educator and philosopher. Best known for *Pedagogy of the Oppressed*
- Argued that education is never neutral – teaching can either reproduce inequality or challenge it
- Major influence on critical pedagogy – education more than learning information. A struggle over voice, power, and human agency.



Socio-Political Factors

“Political” → Politics as in “the struggle for control and influence” or “power dynamics”

- Learning happens through dialogue and inquiry, not through the transfer of information.
- Learners are not deficient containers but active knowers.
- Power shapes who is recognized as capable of knowing, and what is worth learning.

Discuss: Whose / What Forms of Knowledge Are Valued

- What forms of knowledge are valued more by society? Why?
- See if your group can think of examples of knowledge that is seen as:
 - **Mandatory:** “everyone needs to know X”
 - **Optional:** “nice to know, but not necessary”
 - **Coercive:** if you cover this, you are indoctrinating.”
- Why?

Example: Social Group Stereotypes

- What is already presumed about the learner and what they can do / know.
- Stereotypes can influence student performance (e.g., [stereotype threat](#)) – powered by media and social media algorithms
- Different identities can get “steered” to different learning experiences – which can impact ideas around self-efficacy (what students feel like they’re capable of).

Can you think of a time when someone talked at you in a way that revealed assumptions they were making about you?

What We've Learned So Far

- **Biology** helps explain physical learning processes (neurons, plasticity, pathways, input / output).
- **Psychology** helps explain cognitive (Skinner, Piaget) and affective dimensions of behavior, motivation, and learning.
- **Social and political analysis** shows that learning is also collective, cultural, and unequal, and shaped by many external factors.

Each lens reveals something different.

So what? What does this have to do with AI?

- AI is often described as if learning were only an individual, technical process.
- But human learning is social, cultural, and shaped by tools.
- AI systems are also built, trained, and used inside social worlds.

“What an AI model can do?” → “What forms of human learning, judgment, and support does it replace, reshape, or depend on?”

Transition to Career Discussion

- Career development is also social.
- People do not figure out the future alone.
- Values, support, coaching, and networks shape what feels possible.
- We're going to use that idea in today's career work.

Career Discussion

Where you are with your decision about what to do after graduation?

- **Fully undecided:** you have not decided what you would like to do with your degree after graduation
- **Decided, but not taken action:** you know what you would like to do with their degree after graduation, but have not taken steps to pursue it
- **Decided and taken action:** you know what you would like to do with your degree after graduation, and have taken at least one step toward pursuing it